

# Supply Chain Analytics & Performance Optimization

## Project Overview:

*This project focuses on analyzing supply chain operations using SQL and Python to uncover key business insights related to revenue performance, warehouse efficiency, inventory management, lead time, and order fulfillment. The analysis was conducted on 9,000 supply chain records and involved data validation, exploratory data analysis (EDA), and data visualization.*

*The objective of this project is to identify operational trends, evaluate supply chain performance, and support data-driven decision-making through meaningful visual insights. Various analytical techniques were applied to assess revenue patterns, product performance, warehouse operations, and delivery efficiency.*

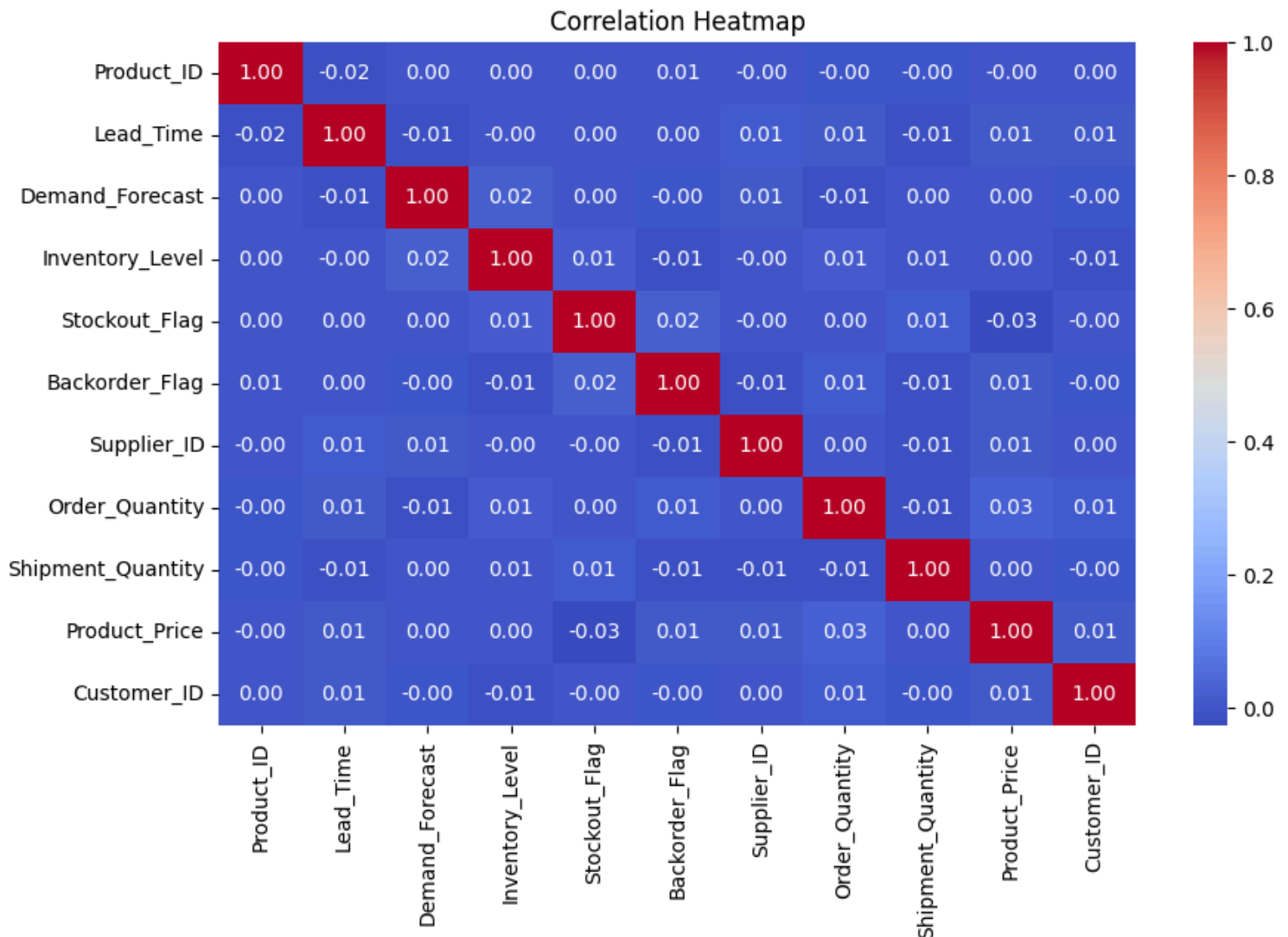
## Tools & Technologies

- SQL (Data Validation & Analysis)
- Excel
- Python
- Pandas
- Matplotlib
- Seaborn
- Power BI

## Dataset Size

- 9,000 Records
- Multiple Warehouses
- Product Categories
- Inventory & Demand Metrics
- Lead Time & Order Information

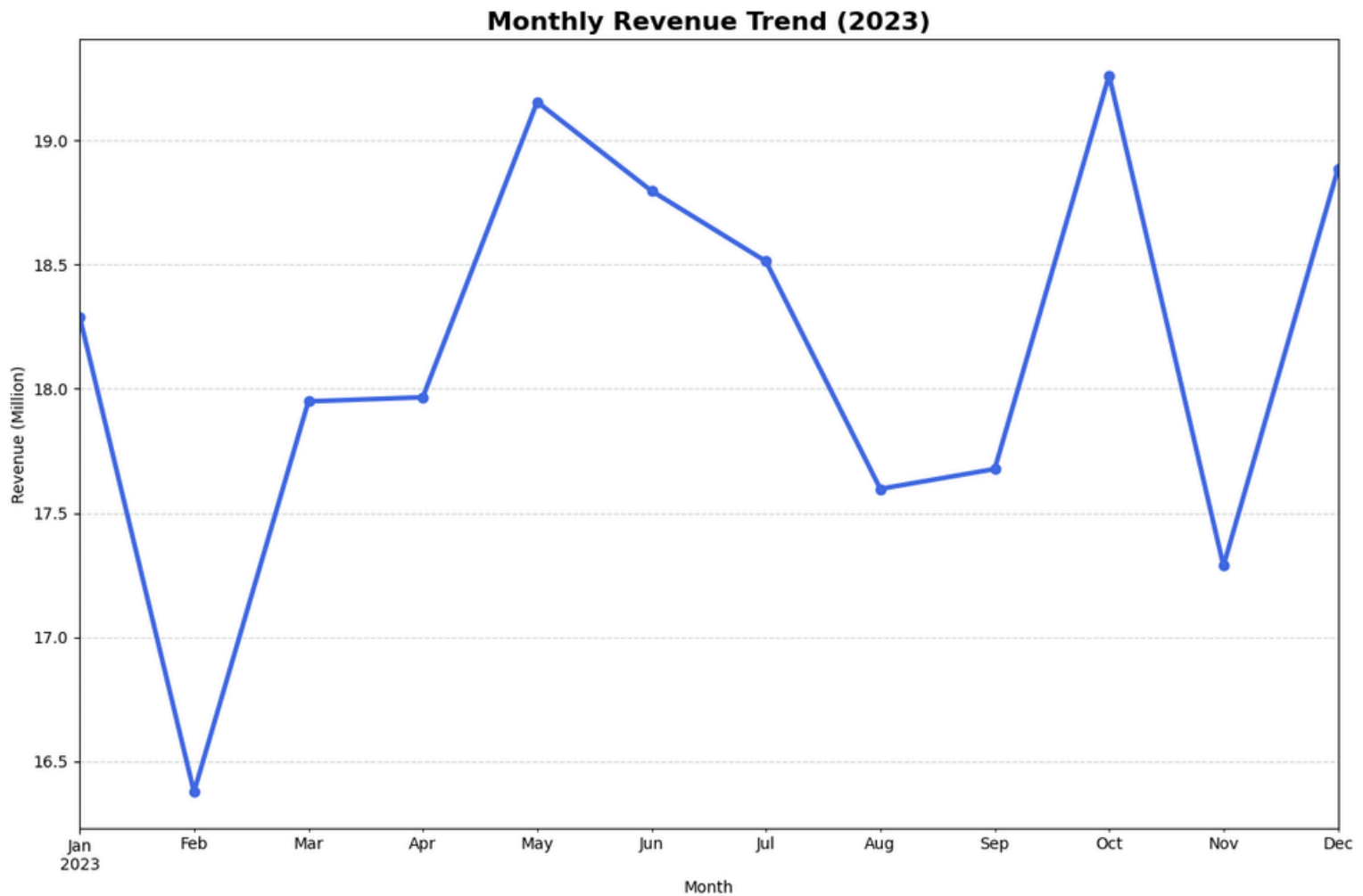
# Heatmap of data



## Correlation Analysis Insights

- The correlation heatmap indicates that most variables have very weak correlations with each other, with correlation coefficients close to zero.
- Demand Forecast and Inventory Level show minimal correlation, suggesting that inventory levels are not strongly driven by forecasted demand in the dataset.
- Lead Time exhibits negligible relationships with operational metrics such as stockouts and backorders, indicating no significant dependency.
- Order Quantity and Shipment Quantity show only a weak positive relationship, suggesting relatively independent variation across orders.
- No strong positive or negative correlations were identified among the analyzed variables, indicating that multiple factors may influence supply chain performance rather than a single dominant metric.
- The absence of strong correlations suggests the need for deeper analysis using segmentation, trend analysis, and predictive modeling rather than relying solely on linear relationships.

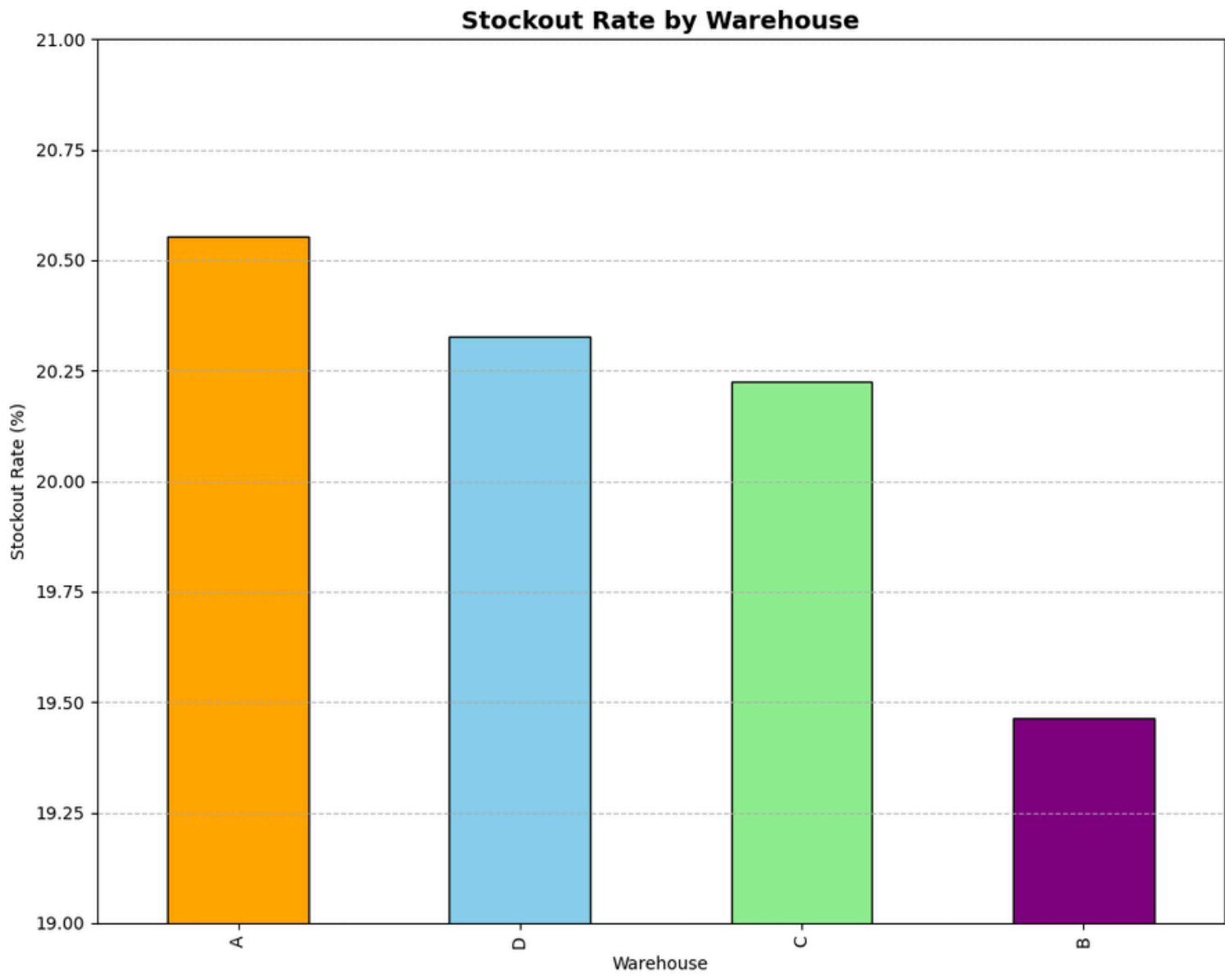
# Monthly Revenue Performance Analysis (2023)



## Key Insights

- Revenue remained relatively stable throughout 2023, ranging between approximately 16.4M and 19.3M.
- The highest monthly revenue was recorded in October 2023, indicating a period of strong sales performance and operational efficiency.
- February 2023 experienced the lowest revenue, followed by a steady recovery during the subsequent months.
- Revenue showed consistent growth from March to May, reaching one of the highest levels during the year.
- Despite minor fluctuations, the overall revenue trend remained stable, demonstrating consistent business performance across the year.
- The strong recovery observed in December suggests increased demand and improved revenue generation towards the end of the year.

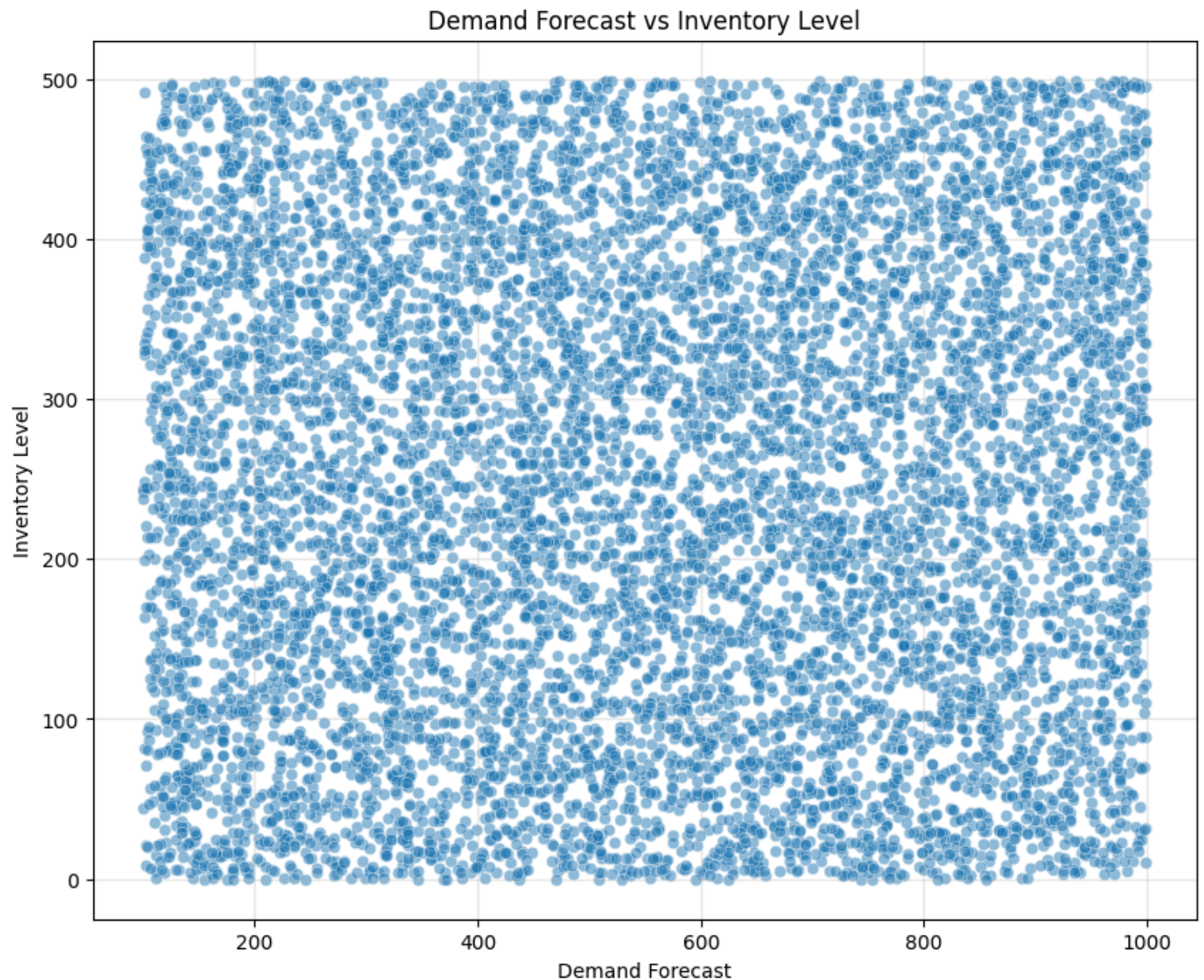
# Warehouse Stockout Rate Analysis



## Key Insights

- Warehouse A recorded the highest stockout rate at approximately 20.5%, indicating a relatively higher risk of inventory shortages.
- Warehouse B achieved the lowest stockout rate at approximately 19.5%, demonstrating comparatively better inventory availability.
- Stockout rates across all warehouses remained within a narrow range of 19.5%–20.5%, suggesting consistent inventory management practices across the supply chain network.
- The small variation between warehouses indicates that stockout challenges are not isolated to a single location and may require network-wide inventory optimization.
- Continuous monitoring of inventory levels and demand forecasting can help further reduce stockout occurrences and improve order fulfillment performance.

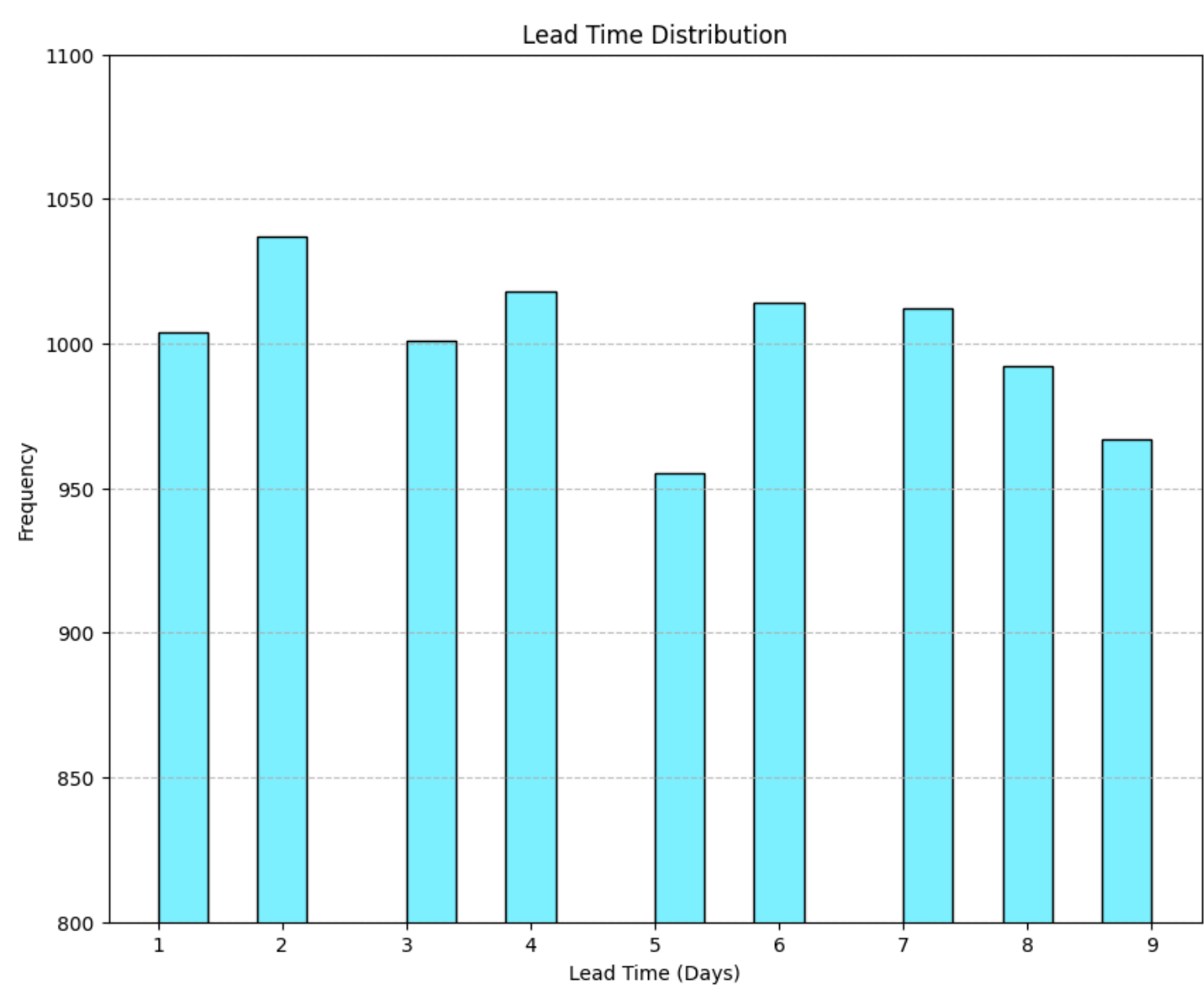
# Demand Forecast vs Inventory Level Analysis



## Key Insights

- The scatter plot shows a wide distribution of inventory levels across different demand forecasts.
- No strong linear relationship is observed between Demand Forecast and Inventory Level, indicating that inventory levels do not consistently increase with forecasted demand.
- Inventory is distributed across both low-demand and high-demand products, suggesting a diversified inventory management strategy.
- The absence of clear clustering patterns indicates that multiple operational factors may influence inventory decisions beyond demand forecasts alone.
- Further analysis using product categories, warehouses, or suppliers may provide deeper insights into inventory planning and stock allocation.

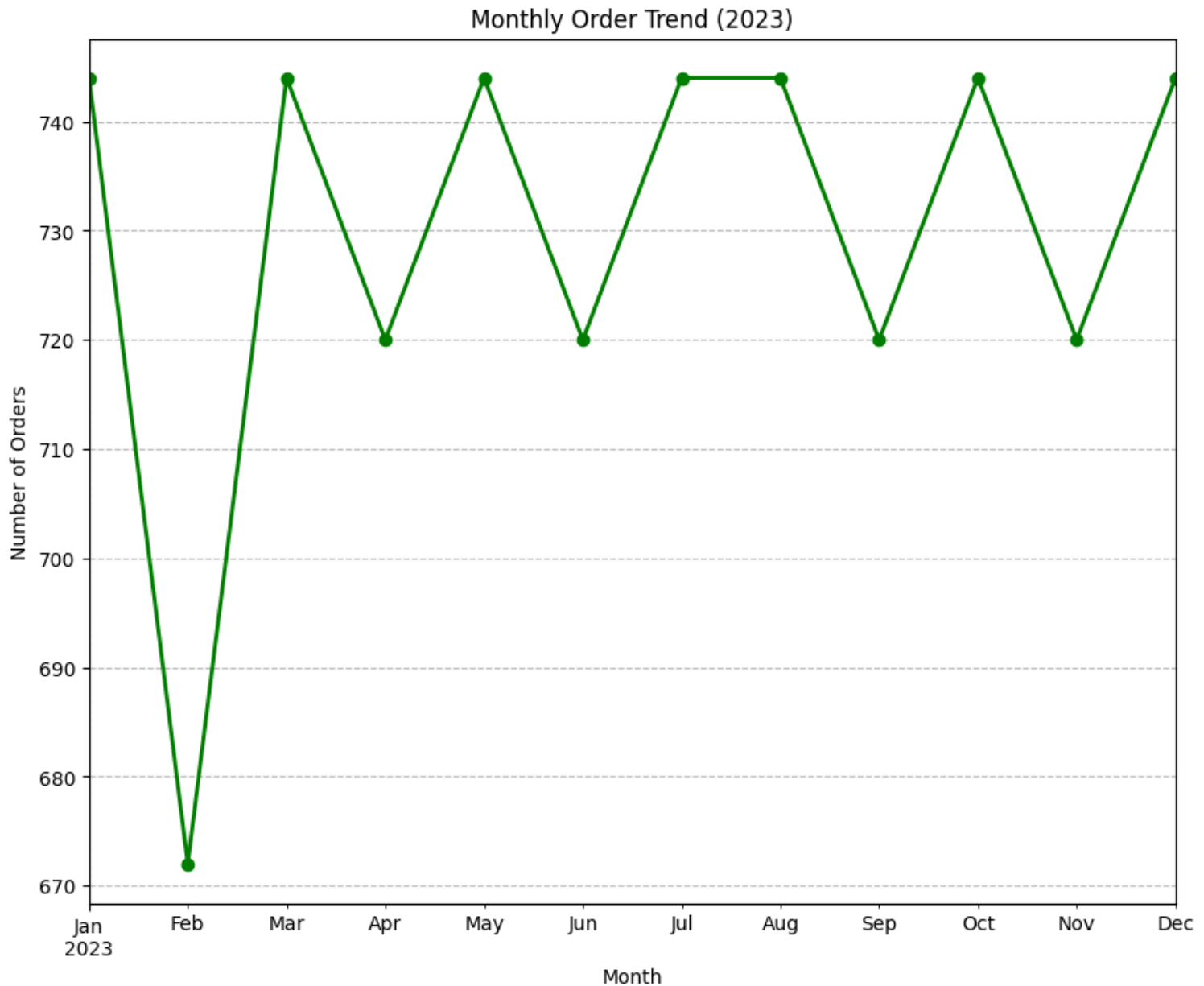
# Distribution of Product Lead Times



## Key Insights

- Lead times are distributed relatively evenly across the 1–9 day range, indicating a balanced supply chain delivery pattern.
- No single lead time dominates the dataset, suggesting that orders are processed and delivered across multiple time windows rather than concentrating on a specific duration.
- The highest order frequency is observed around the 2–4 day lead time range, highlighting efficient fulfillment for a significant portion of orders.
- Variations between lead time categories are relatively small, indicating consistency in supplier and logistics performance.
- The absence of extreme lead time spikes suggests stable operational processes with limited delivery bottlenecks.
- Maintaining lower lead times can further improve customer satisfaction, inventory turnover, and overall supply chain efficiency.

# Monthly Order Trend Analysis (2023)



## Key Insights

- Monthly order volume remained highly stable throughout 2023, averaging approximately 720–745 orders per month.
- February recorded the lowest number of orders, indicating a temporary decline in demand compared to other months.
- Order volume quickly recovered after February and remained consistent for the remainder of the year.
- Peak order activity was observed in multiple months, including January, March, May, July, August, October, and December.
- The limited fluctuation in monthly orders suggests predictable customer demand and stable business operations.
- Consistent order volumes throughout the year indicate effective supply chain planning and balanced operational capacity.